

Chemistry of a Printable Asymmetric MXene Hybrid Lithium-Ion Cell: Hosts, Salt, Separator, and Lithium Inventory

Vincent Mark Garrett
Independent Researcher, Lubbock, Texas, USA

September 23, 2026

Abstract

This paper describes the electrochemistry of one printable lithium-ion cell built from materials that already exist in the literature: a $\text{Ti}_3\text{C}_2\text{T}_x/\text{Nb}_2\text{CT}_x$ negative electrode, a $\text{V}_2\text{O}_5/\text{TiO}_2@\text{Ti}_3\text{C}_2\text{T}_x$ positive electrode, and a cross-linked PVDF–PEO membrane containing LiTFSI, cubic LLZO (or Ta-doped LLZTO), and a few weight percent of MXene. The working ion is Li^+ . The two electrodes occupy different potential windows versus Li/Li^+ . Lithium that is not present in the as-printed hosts is supplied by prelithiation or by a sacrificial source. The separator is kept below the electronic percolation threshold of MXene.

The text is a materials description. Capacities, voltages, and conductivities are those reported for the individual components in half-cells or standalone membranes. They are not scaled to a claimed packaged cell. Synthesis routes are given at the level needed to prepare the powders, inks, and film.

Keywords: $\text{Ti}_3\text{C}_2\text{T}_x$; Nb_2CT_x ; V_2O_5 ; TiO_2 ; LiTFSI; LLZO; PVDF; PEO; prelithiation

1 Cell chemistry in one paragraph

Li^+ is extracted from a prelithiated MXene negative electrode (or from a sacrificial lithium source) on charge and intercalates into V_2O_5 (with a TiO_2 and MXene support) on the positive side. On discharge the ions return. Electrons travel in the metallic MXene sheets and the nanoribbon network. Ions travel in the polymer–salt phase and along LLZO particle surfaces and polymer–ceramic interfaces. The separator must not carry electrons from one foil to the other.

Nominal half-cell windows used to set cutoffs:

- negative: approximately 0.2–2.2 V vs. Li/Li^+ ;
- positive: approximately 2.0–3.8 V vs. Li/Li^+ ;
- implied full-cell window after matching: approximately 1.2–2.6 V, with an average discharge voltage near 1.8–2.4 V.

2 Negative electrode: $\text{Ti}_3\text{C}_2\text{T}_x$ and Nb_2CT_x

2.1 Structure and Li storage

$\text{Ti}_3\text{C}_2\text{T}_x$ is a two-dimensional titanium carbide with surface terminations $\text{T}_x = \text{O}, \text{OH}, \text{F}$ (and sometimes Cl). Lithium occupies interlayer sites and coordinates to surface oxygen. The theoretical capacity of bare Ti_3C_2 is about 320 mAh g^{-1} for $\text{Ti}_3\text{C}_2\text{Li}_2$. Terminations reduce that value: computational work gives roughly 268 mAh g^{-1} for $-\text{O}$, 130 mAh g^{-1} for $-\text{F}$, and 67 mAh g^{-1} for $-\text{OH}$ [1]. Experimentally, freestanding or slurry $\text{Ti}_3\text{C}_2\text{T}_x$ anodes in liquid carbonate electrolyte typically deliver $100\text{--}250 \text{ mAh g}^{-1}$ reversible capacity between about 0.05 and 2.5 V vs. Li/Li^+ , with first-cycle loss of tens of percent from SEI and irreversible trapping on defect and $-\text{OH}/-\text{F}$ sites [2, 6]. Rate capability is high because Li diffusion barriers on O-terminated surfaces are low and the sheets are metallic.

Nb_2CT_x is the $n = 1$ niobium carbide MXene obtained by etching Nb_2AlC . Its theoretical Li capacity is higher than that of $\text{Ti}_3\text{C}_2\text{T}_x$ in several models (values near 500 mAh g^{-1} appear for idealized surfaces). The first experimental report gave about 170 mAh g^{-1} at 1C and 110 mAh g^{-1} at 10C [3]. Later surface-group control (Li pre-intercalation plus annealing to enrich $-\text{O}$ and strip $-\text{F}/-\text{OH}$) raised reversible capacity to about 448 mAh g^{-1} at 0.05 A g^{-1} , with 342 mAh g^{-1} remaining after 2000 cycles at 2 A g^{-1} [7]. Nb_2CT_x restacks readily; spacers or quantum-dot pillars are used to keep galleries open.

The two MXenes are mixed so that (i) $\text{Ti}_3\text{C}_2\text{T}_x$ supplies electronic conductivity and a mature ink literature, and (ii) Nb_2CT_x supplies additional interlayer capacity and a second c -lattice spacing. They occupy the same potential window. The mixture is an anode, not a cathode.

2.2 Recommended dry composition

Component	wt% of dry electrode
Delaminated $\text{Ti}_3\text{C}_2\text{T}_x$	55
Delaminated Nb_2CT_x	25
N-doped graphene nanoribbons (4–6 at.% N)	8
PVDF (or PVDF–PEO) binder	12

Target after drying and light calendaring: $2\text{--}4 \text{ mg cm}^{-2}$ active mass, 30–40% porosity, 30–50 μm thickness. Design capacity for matching: 180 mAh g^{-1} of active mass at low rate, until measured.

2.3 Synthesis of $\text{Ti}_3\text{C}_2\text{T}_x$ (MILD)

The minimally intensive layer delamination (MILD) route avoids concentrated HF as the poured reagent and yields larger flakes than harsh sonication of HF-etched multilayer powder [4, 5].

Typical laboratory batch for 1 g Ti_3AlC_2 ($< 38 \mu\text{m}$):

1. Dissolve 1.6 g LiF in 20 mL of 9 M HCl in a polypropylene bottle. Stir until clear. This generates HF *in situ* and provides Li^+ for intercalation.
2. Add 1.0 g Ti_3AlC_2 slowly (exotherm). Molar LiF:MAX is about 7.5:1.

3. Stir 24 h at 35 °C.
4. Wash by repeated centrifugation with deionized water until the supernatant pH is 5–6.
5. Collect the swollen clay. Delaminate by hand-shaking or brief ice-bath sonication under Ar. Centrifuge at low speed to drop unetched MAX; collect the dark-green supernatant.
6. Store as a concentrated aqueous colloid under Ar at 4 °C, or freeze-dry. Do not dry in air at elevated temperature if oxidation must be avoided.

Quality checks: XRD should show loss of the MAX (104) peak near 39° and a shift of the (002) peak to lower angle ($2\theta \approx 6\text{--}8^\circ$). XPS should show little residual Al. Aqueous dispersions oxidize; use promptly or keep oxygen-free.

Fluorine-lean, oxygen-rich terminations improve Li capacity. Alkaline wash or controlled annealing after Li intercalation can shift T_x toward $-\text{O}$. That step must be validated by XPS before it is treated as part of the baseline recipe.

2.4 Synthesis of Nb_2CT_x

Nb_2AlC is etched analogously. Nb_2AlC etches more slowly than Ti_3AlC_2 ; literature protocols use LiF/HCl or HF for 48–96 h, sometimes at slightly elevated temperature. After washing, LiOH soak plus annealing near 400 °C has been used to enrich $-\text{O}$ and raise capacity [7]. Delaminated few-layer Nb_2CT_x is the form needed for inks. Multilayer accordion powder without delamination will under-perform.

2.5 Nitrogen-doped graphene nanoribbons

Multi-walled carbon nanotubes are oxidatively unzipped (modified Hummers or $\text{KMnO}_4/\text{H}_2\text{SO}_4$), washed, and hydrothermally treated with urea to introduce 4–6 at.% N. The ribbons sit between MXene sheets, reduce restacking, and add an electronic path that survives if some MXene oxidizes to TiO_2 . Eight weight percent is a spacer level, not a primary host.

2.6 Negative-electrode ink

Disperse the two MXenes and the nanoribbons in water or NMP/water (70:30) with the binder. Target viscosity for slot-die is several thousand cP; aerosol-jet inks are more dilute and must be stable against oxidation for hours. Degassing and ice-bath sonication (short) help. Zeta potential of MXene colloids is typically negative; avoid pH swings that collapse the double layer.

3 Positive electrode: $\text{V}_2\text{O}_5/\text{TiO}_2@\text{Ti}_3\text{C}_2\text{T}_x$

3.1 Why this hybrid

Orthorhombic V_2O_5 is a layered oxide that intercalates Li over several phase sequence ($\alpha \rightarrow \epsilon \rightarrow \delta \rightarrow \gamma$, and further to rock-salt-like ω if driven below $\sim 2\text{ V}$). Theoretical capacity is 147 mAh g^{-1} for one Li per V_2O_5 and 294 mAh g^{-1} for two. Deeper lithiation gives higher capacity at the

cost of structural damage. Electronic conductivity of V_2O_5 is poor. Anchoring V_2O_5 (and adventitious or grown TiO_2) on $\text{Ti}_3\text{C}_2\text{T}_x$ shortens electron paths and limits agglomeration.

Myint et al. prepared a $\text{V}_2\text{O}_5/\text{TiO}_2@\text{Ti}_3\text{C}_2$ hybrid by solution mixing of a vanadyl precursor with $\text{Ti}_3\text{C}_2\text{T}_x$ and thermal decomposition [8]. The optimized composition with about 5 wt% MXene in the hybrid (VM5) delivered about 460 mAh g^{-1} at 0.1 A g^{-1} and 290 mAh g^{-1} at 1 A g^{-1} versus Li metal, with 62% retention over 500 cycles at 1 A g^{-1} . Charge storage was mixed: roughly three-quarters capacitive and one-quarter diffusion-controlled at 0.4 mV s^{-1} . First-cycle irreversible capacity was large when the electrode was taken down to 0.01 V. For a positive electrode opposite an MXene anode, the voltage window should stay in the intercalation region (about 2.0–3.8 V vs. Li/Li⁺), not the conversion-like deep-discharge region.

Design capacity for matching: 200 mAh g^{-1} of hybrid mass in the 2.0–3.8 V window, until measured in the printed film.

3.2 Recommended dry composition

Component	wt% of dry electrode
$\text{V}_2\text{O}_5/\text{TiO}_2@\text{Ti}_3\text{C}_2\text{T}_x$ hybrid	70
Additional $\text{Ti}_3\text{C}_2\text{T}_x$ or carbon (if hybrid is MXene-poor)	10
N-doped graphene nanoribbons	8
PVDF binder	12

The MXene fraction *inside the hybrid* should remain low (5–15 wt% of the hybrid), as in the reported VM5 material. Raising MXene in the hybrid increases conductivity but dilutes the vanadium inventory.

Target loading: capacity-matched to the negative electrode after first-cycle loss, typically 3–6 mg cm^{-2} hybrid, 40–60 μm dry thickness.

3.3 Laboratory synthesis of the hybrid

A procedure aligned with [8]:

1. Disperse washed $\text{Ti}_3\text{C}_2\text{T}_x$ in water or ethanol under Ar.
2. Add a vanadium source (vanadyl acetylacetonate or vanadyl sulfate) and, if a distinct TiO_2 shell is required beyond MXene oxidation, a titanium alkoxide (titanium isopropoxide) at a controlled V:Ti ratio.
3. Stir, then dry. Alternatively, treat hydrothermally at 180 °C for 8–12 h.
4. Calcination in air at a mild temperature converts the vanadium precursor to orthorhombic V_2O_5 and grows nanocrystalline TiO_2 on residual carbide. Excessive air calcination destroys the MXene core; XRD and XPS must show remaining carbide C–Ti intensity if the metallic core is intended.
5. Confirm phases: XRD peaks of $\alpha\text{-V}_2\text{O}_5$, anatase or rutile TiO_2 , and residual MXene (002). TEM should show oxide on sheets, not isolated micron-scale V_2O_5 only.

V_2O_5 dissolves in some aqueous and acidic media and can disproportionate. Inks for the positive electrode are better made in NMP with PVDF than in water.

4 Lithium inventory

As-printed $Ti_3C_2T_x$, Nb_2CT_x , and V_2O_5 contain negligible extractable Li. LiTFSI in a 20–40 μm membrane is a few tens of $\mu Ah\ cm^{-2}$ at most—not enough for a $0.5\ mAh\ cm^{-2}$ cell after SEI formation.

Three workable sources:

1. **Electrochemical prelithiation of the negative electrode.** Assemble a temporary Li — liquid electrolyte — printed MXene half-cell, lithiate to a chosen cutoff (e.g. 0.2 V vs. Li/Li⁺), disassemble in the dry room, rinse, dry, and laminate against the positive electrode and solid membrane. Capacity of prelithiation must cover full-cell areal capacity plus 20–40% first-cycle loss.
2. **Chemical prelithiation.** Stabilized lithium-metal powder or a molecular lithiation reagent contacted to the MXene film. Residual reagent must be removed; this is harder to control than electrochemical prelithiation.
3. **Sacrificial lithium additive in the positive electrode** (e.g. a Li_5FeO_4 -type or Li_2O/Co sacrificial component) that releases Li⁺ on first charge and is not required thereafter. This changes the positive-electrode mass balance and must be counted in the bill of materials.

Option 1 is the baseline for a first prototype. The paper does not describe a lithium-free cell.

5 Separator: PVDF–PEO–LiTFSI–LLZO with dilute MXene

5.1 Roles

- PEO ether oxygens solvate Li⁺ and provide segmental-motion transport above the glass transition and in amorphous regions.
- PVDF supplies mechanical integrity and a higher melting range than PEO alone. A 60:40 to 70:30 PVDF:PEO mass ratio is a practical window for free-standing films [10].
- LiTFSI is the salt. Bis(trifluoromethanesulfonyl)imide dissociates readily in ethers; EO:Li between 12:1 and 20:1 is the usual starting range. Lower EO:Li raises carrier concentration and then raises viscosity and ion pairing.
- Cubic LLZO / Ta-doped LLZTO particles add a ceramic Li⁺ path, scatter PEO crystallites, and stiffen the film. Ten to fifteen weight percent is enough to matter and low enough to cast.
- Two to five weight percent $Ti_3C_2T_x$ further suppresses PEO crystallization and provides surface sites that interact with Li⁺ [9]. Above that loading, MXene forms an electronic network. Alignment of sheets through the film thickness is not used.

5.2 Recommended dry membrane composition

Component	wt% of dry membrane
PVDF + PEO (60:40 by mass)	72
LiTFSI (set by EO:Li = 16)	counted inside the polymer-salt fraction
Ta-doped LLZTO (e.g. $\text{Li}_{6.4}\text{La}_3\text{Zr}_{1.6}\text{Ta}_{0.4}\text{O}_{12}$)	12
$\text{Ti}_3\text{C}_2\text{T}_x$	3
Thermal or peroxide cross-linker residue	balance (~ 1 – 2)

LiTFSI mass is fixed by the PEO content and the chosen EO:Li ratio, not by a free weight percent. For PEO with repeat-unit mass 44 g mol^{-1} and LiTFSI mass 287 g mol^{-1} , EO:Li = 16 corresponds to $m_{\text{LiTFSI}}/m_{\text{PEO}} = 287/(16 \times 44) \approx 0.41$.

Target thickness 20–40 μm . Target $\sigma_{\text{ion}}(25^\circ\text{C})$ of order $10^{-4} \text{ S cm}^{-1}$. Electronic conductivity by DC polarization must be $\leq 10^{-9} \text{ S cm}^{-1}$. If it is not, the MXene loading is reduced until it is.

Optional plasticizer (succinonitrile or a low-vapor-pressure carbonate, $\leq 10 \text{ wt}\%$) is a separate decision. It raises conductivity and makes the membrane quasi-solid. If used, the paper must say so.

5.3 LLZTO powder

Solid-state route (representative):

1. Mix $\text{LiOH}\cdot\text{H}_2\text{O}$ or Li_2CO_3 (10–15 mol% excess Li), La_2O_3 (pre-dried), ZrO_2 , and Ta_2O_5 in the $\text{Li}_{6.4}\text{La}_3\text{Zr}_{1.6}\text{Ta}_{0.4}\text{O}_{12}$ stoichiometry.
2. Ball-mill in isopropanol, dry, calcine near 900°C , mill again.
3. Fire 1100–1175 $^\circ\text{C}$ in a MgO crucible under mother powder to limit Li loss and obtain cubic garnet.
4. Mill to sub-micron particles. Surface-modify with a silane (APTES) or a short PEO brush so the particles wet the polymer solution and do not settle as bare ceramic.

XRD must show cubic garnet without dominant $\text{La}_2\text{Zr}_2\text{O}_7$ pyrochlore. Exposed LLZO reacts with CO_2 and H_2O to form Li_2CO_3 ; store dry.

5.4 Membrane casting

In a dry room or Ar glovebox:

1. Dissolve PVDF and PEO in anhydrous DMF or NMP at 60–80 $^\circ\text{C}$.
2. Add LiTFSI; stir until clear.
3. Disperse LLZTO and MXene separately (brief sonication), then combine. Avoid long sonication of MXene in the salt solution.

4. Add a thermal cross-linker (e.g. a small fraction of peroxide or a multifunctional acrylate compatible with PEO) if a covalently cross-linked network is required.
5. Cast or slot-die. Dry under staged vacuum at 40–80 °C to remove solvent. Residual DMF coordinated to Li^+ is a known poison for PVDF electrolytes and must be driven out.
6. Optional light cross-link at 80–120 °C. Do not heat the salted membrane to 410 °C.

6 Current collectors and contacts

Positive: 8–12 μm Al foil. Negative: Al, carbon-coated Al, or a printed carbon film. A dilute MXene dip-coat on the foil lowers contact resistance. Check Al alloying and corrosion on the negative side after cycling; switch to carbon-coated foil if needed.

7 Interfaces

Three interfaces dominate impedance.

1. **MXene — polymer.** Terminations ($-\text{OH}$, $-\text{O}$, $-\text{F}$) hydrogen-bond to PEO and PVDF. That contact is chemically plausible. The risk is MXene oxidation in residual moisture, which grows insulating TiO_2 and consumes Li.
2. **V_2O_5 — polymer.** PEO is not stable at high potential. The positive-electrode cutoff should stay near or below 3.8 V vs. Li/Li^+ until oxidative stability of the specific membrane is measured by LSV on inert electrodes and on the hybrid cathode.
3. **LLZTO — polymer.** Unmodified garnet forms Li_2CO_3 skins and high interfacial resistance. Silane or polymer-brush modification is part of the recipe, not an optional extra.

8 Assembly and formation

Print or coat the negative electrode on its foil; dry 80–120 °C under inert gas. Print or coat the positive electrode; dry similarly. Prelithiate the negative electrode. Lay in the membrane. Laminate at 0.5–1.0 MPa and 70–90 °C. Pouch in a dry room.

Formation: C/20 to C/10 for two to five cycles with voltage limits taken from the half-cell windows after iR correction. Record first-cycle loss. Subsequent cycling at C/5 to 1C is the characterization range. Higher rates are measurements, not specifications.

9 What the literature supports, and what it does not

Supported as individual components: MILD $\text{Ti}_3\text{C}_2\text{T}_x$ inks [11]; Nb_2CT_x as a Li host [3, 7]; $\text{V}_2\text{O}_5/\text{TiO}_2@/\text{Ti}_3\text{C}_2$ half-cell behavior [8]; PEO–LiTFSI with a few percent MXene [9]; PVDF–PEO–Li-salt films [10]; LLZO-filled PVDF-family membranes.

Not supported by those same papers: a 3 V symmetric MXene cell; a separator that is one-third MXene by mass; cell-level energy density of 400 Wh kg^{-1} from these hosts at 150 μm

with metal foils; 10C operation of a polymer cell at high areal loading; 10^5 cycles of the V_2O_5 hybrid in a solid full cell.

A stack-level estimate using 180 mAh g^{-1} (negative active), 200 mAh g^{-1} (positive hybrid), 2.1 V average, and realistic foil plus membrane mass falls in a broad band around 80–150 Wh kg^{-1} before pouch packaging. That is the energy scale this chemistry occupies.

10 Characterization that defines the chemistry

Before any rate table is written, measure:

1. XRD, XPS, and Raman of each powder (phase purity, T_x chemistry, residual Al).
2. Half-cell GCD and CV of each printed electrode vs. Li in liquid electrolyte, then vs. the solid membrane.
3. Separator σ_{ion} (EIS, 25 and 60 °C) and σ_e (DC polarization).
4. Li^+ transference number of the membrane.
5. Oxidative stability (LSV) to at least 4.2 V vs. Li/Li $^+$.
6. Full-cell first-cycle loss after the stated prelithiation protocol.
7. Post-mortem XRD/XPS of both electrodes after 50 cycles.

11 Conclusion

The surviving chemistry is a rocking-chair lithium-ion cell. The negative host is a $Ti_3C_2T_x/Nb_2CT_x$ mixed MXene with a nanoribbon spacer. The positive host is a V_2O_5 -rich oxide grown on a minority MXene support. The ion conductor is a PVDF–PEO–LiTFSI matrix with a moderate LLZO loading and a MXene loading low enough that the film remains an electronic insulator. Lithium is introduced on purpose. Process heat stays inside the polymer window. That is the full materials specification; performance beyond the component literature is a measurement, not part of the specification.

References

- [1] Q. Tang, Z. Zhou, P. Shen, *J. Am. Chem. Soc.* **134**, 16909 (2012).
- [2] M. Naguib et al., *Electrochem. Commun.* **16**, 61 (2012).
- [3] M. Naguib et al., *J. Am. Chem. Soc.* **135**, 15966 (2013).
- [4] M. Alhabeb et al., *Chem. Mater.* **29**, 7633 (2017).
- [5] M. Ghidui, M. R. Lukatskaya, M.-Q. Zhao, Y. Gogotsi, M. W. Barsoum, *Nature* **516**, 78 (2014).

- [6] *ACS Omega* **7**, 40682 (2022).
- [7] J. Zhao et al., *J. Energy Chem.* **53**, 387 (2021).
- [8] W. Myint et al., *ACS Appl. Mater. Interfaces* **16**, 53764 (2024).
- [9] Q. Pan et al., *Nanoscale Adv.* **1**, 395 (2019).
- [10] Free-standing Li⁺-conductive films based on PEO–PVDF blends, *RSC Adv.* (2020), DOI: 10.1039/D0RA02325F.
- [11] C. J. Zhang et al., *Nat. Commun.* **10**, 1795 (2019).